L&B Problem 2: The Real Scalar Field

Mode Expansion, Lorentz-Invariant Normalization,
and the Road to the Hamiltonian

Based on Lancaster & Blundell, *Quantum Field Theory for the Gifted Amateur*
Chapters 6 and 11

The Big Picture: What Are We Doing?

In classical mechanics a field $\phi(\mathbf{x}, t)$ assigns a number to every point in space and time. Quantizing it means promoting ϕ to an *operator* that can create and destroy quanta (particles). The central tool is the **mode expansion**: we decompose the field operator into a sum (integral) over momentum modes, each weighted by a creation or annihilation operator.

The three parts of this problem trace a single logical arc:

(a) *Write down* the mode expansion and understand its two terms.

(b) *Explain* the mysterious $1/\sqrt{2E}$ factor that appears in front.

(c) *Track* how factors of E propagate through the derivation of the Hamiltonian, ending at Eq. 11.38.

Part (a): The Mode Expansion of the Real Scalar Field

Reminder — the single harmonic oscillator (Eq. 6.19)

Before tackling the field, recall the *quantum harmonic oscillator*. In Chapter 6, L&B write the position operator of a single oscillator with frequency ω as

Equation 6.19 — Single Oscillator

$$\hat{q} = \frac{1}{\sqrt{2\omega}}(\hat{a} + \hat{a}^\dagger),$$

where $\hat{a}$ and $\hat{a}^\dagger$ satisfy $[\hat{a}, \hat{a}^\dagger] = 1$.

What the two pieces mean

- $\hat{a}$ — the **annihilation (lowering) operator**. Acting on the n-th energy eigenstate it removes one quantum: $\hat{a}\,|n\rangle = \sqrt{n}\;|n-1\rangle$.
- $\hat{a}^\dagger$ — the **creation (raising) operator**. It adds one quantum: $\hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\;|n+1\rangle$.

The factor $1/\sqrt{2\omega}$ sets the *amplitude scale* so that $\hat{q}$ has the correct dimensions and the Hamiltonian $\hat{H} = \omega(\hat{a}^\dagger\hat{a} + \frac{1}{2})$ is recovered.

Promoting to a field — the mode expansion (Chapter 11)

A real scalar field is, conceptually, an *infinite collection of harmonic oscillators*, one for each momentum $\mathbf{p}$. The oscillator labeled by $\mathbf{p}$ has frequency $\omega_{\mathbf{p}} \equiv E_{\mathbf{p}} = \sqrt{|\mathbf{p}|^2 + m^2}$ (we use natural units $\hbar = c = 1$).

Each oscillator gets its own pair of ladder operators $\hat{a}_{\mathbf{p}}$, $\hat{a}^\dagger_{\mathbf{p}}$. Summing (integrating) over all momenta gives the field operator:

Mode Expansion of the Real Scalar Field (Chapter 11)

$$\hat{\phi}(x) \;=\; \int \frac{d^3p}{(2\pi)^3}\,\frac{1}{\sqrt{2E_{\mathbf{p}}}}\Big[\,\hat{a}_{\mathbf{p}}\,e^{ip\cdot x} \;+\; \hat{a}^\dagger_{\mathbf{p}}\,e^{-ip\cdot x}\Big] \tag{1}$$

where $p\cdot x = E_{\mathbf{p}}\,t - \mathbf{p}\cdot\mathbf{x}$, and $[\hat{a}_{\mathbf{p}},\,\hat{a}^\dagger_{\mathbf{q}}] = (2\pi)^3\,\delta^{(3)}(\mathbf{p}-\mathbf{q})$.

Connection to Eq. 6.19

From one oscillator to infinitely many

Equation 6.19 has the structure $\hat{q} = \frac{1}{\sqrt{2\omega}}(\hat{a} + \hat{a}^\dagger)$. The mode expansion (1) is its direct generalization:

- The single frequency ω is replaced by the continuum label $E_{\mathbf{p}} = \sqrt{|\mathbf{p}|^2 + m^2}$.
- The single pair $(\hat{a}, \hat{a}^\dagger)$ becomes a family $(\hat{a}_{\mathbf{p}}, \hat{a}^\dagger_{\mathbf{p}})$, one for each $\mathbf{p}$.
- The sum $(\hat{a} + \hat{a}^\dagger)$ becomes an integral over momenta.
- Plane-wave phases $e^{\pm ip\cdot x}$ appear because each mode oscillates in both space and time.

What the two terms represent:

1. $\hat{a}_{\mathbf{p}}\,e^{ip\cdot x}$ — positive-frequency part: This term *destroys* a particle of momentum $\mathbf{p}$. Its space-time dependence $e^{i(E_{\mathbf{p}}t - \mathbf{p}\cdot\mathbf{x})}$ is a positive-frequency plane wave (energy $E > 0$, propagating forward in time in the Schrödinger picture).
2. $\hat{a}^\dagger_{\mathbf{p}}\,e^{-ip\cdot x}$ — negative-frequency part: This term *creates* a particle of momentum $\mathbf{p}$. Its conjugate phase $e^{-i(E_{\mathbf{p}}t - \mathbf{p}\cdot\mathbf{x})}$ is a negative-frequency plane wave.

The requirement that $\hat{\phi}$ be **Hermitian** (real scalar field $\Rightarrow \hat{\phi} = \hat{\phi}^\dagger$) forces the creation piece to be the Hermitian conjugate of the annihilation piece, linking the two terms.

Part (b): Where Does $1/\sqrt{2E}$ Come From?

The factor $1/\sqrt{2E_{\mathbf{p}}}$ looks innocuous, but it encodes deep physics: **Lorentz invariance**. Lancaster & Blundell build the explanation in three interlocking pieces.

Piece 1: The Lorentz-invariant integration measure

Under a Lorentz boost a spatial volume element d^3p is *not* invariant — it Lorentz-contracts. However, the combination

$$\frac{d^3p}{2E_{\mathbf{p}}}$$

is Lorentz invariant. One way to see this: the four-dimensional integral $\int d^4p\ \delta(p^2 - m^2)\,\theta(p^0)$ is manifestly Lorentz-invariant, and performing the p^0 integral using the delta function produces exactly $\int d^3p/(2E_{\mathbf{p}})$.

Key identity

$$\int \frac{d^4p}{(2\pi)^4}\,(2\pi)\,\delta(p^2 - m^2)\,\theta(p^0) \;=\; \int \frac{d^3p}{(2\pi)^3\,2E_{\mathbf{p}}}$$

This is the *Lorentz-invariant measure on the mass shell*. Every $1/(2E)$ you encounter in QFT traces back to this.

Piece 2: Lorentz-covariant normalization of momentum eigenstates

L&B adopt the **covariant normalization convention**:

$$\langle \mathbf{p}|\mathbf{q}\rangle \;=\; 2E_{\mathbf{p}}\,(2\pi)^3\,\delta^{(3)}(\mathbf{p}-\mathbf{q}).$$

Why the extra $2E_{\mathbf{p}}$? With the non-relativistic convention $\langle \mathbf{p}|\mathbf{q}\rangle = (2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q})$, the normalization is *not* boost-invariant (it picks up a frame-dependent factor). Inserting $2E_{\mathbf{p}}$ exactly compensates the Lorentz contraction of d^3p, so that the completeness relation

$$\mathbf{1} = \int \frac{d^3p}{(2\pi)^3\,2E_{\mathbf{p}}}\ |\mathbf{p}\rangle\,\langle\mathbf{p}|$$

is manifestly Lorentz-invariant.

Connecting normalization to the mode expansion

If we define the single-particle states by

$$|\mathbf{p}\rangle \;=\; \sqrt{2E_{\mathbf{p}}}\,\hat{a}^\dagger_{\mathbf{p}}\,|0\rangle\,,$$

and the ladder operators satisfy $[\hat{a}_{\mathbf{p}}, \hat{a}^\dagger_{\mathbf{q}}] = (2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q})$, then

$$\langle \mathbf{p}|\mathbf{q}\rangle = 2E_{\mathbf{p}}\,(2\pi)^3\,\delta^{(3)}(\mathbf{p}-\mathbf{q}). \quad \checkmark$$

The $\sqrt{2E_{\mathbf{p}}}$ dressing of the creation operator is what forces the $1/\sqrt{2E_{\mathbf{p}}}$ to appear in the mode expansion.

Piece 3: Putting it all together

There are two equivalent ways to see where $1/\sqrt{2E}$ lands:

Viewpoint A — Start from the Lorentz-invariant integral. Write the field as an integral over the invariant measure:

$$\hat{\phi}(x) = \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}} \Big[\underbrace{\sqrt{2E_{\mathbf{p}}}\, \hat{a}_{\mathbf{p}}}_{\text{creates } |\mathbf{p}\rangle/\sqrt{2E_{\mathbf{p}}}} e^{ip\cdot x} + \text{h.c.} \Big].$$

Pulling $\sqrt{2E}$ out of the numerator and combining with $2E$ in the denominator gives $\sqrt{2E}/(2E) = 1/\sqrt{2E}$ in front.

Viewpoint B — Generalize Eq. 6.19 directly. Each mode is a harmonic oscillator with $\omega \to E_{\mathbf{p}}$. The single-oscillator result $1/\sqrt{2\omega}$ immediately becomes $1/\sqrt{2E_{\mathbf{p}}}$.

Both routes land on the same formula, confirming that the $1/\sqrt{2E_{\mathbf{p}}}$ factor simultaneously:

- ensures Lorentz invariance of the field;
- matches the covariant state normalization $\langle \mathbf{p}|\mathbf{q}\rangle = 2E_{\mathbf{p}}(2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q})$;
- is the natural generalization of the $1/\sqrt{2\omega}$ amplitude in Eq. 6.19.

Part (c): Tracking the Factor of E Through to the Hamiltonian

We now trace how energy-dependent factors evolve step-by-step in the derivation of Eq. 11.38. Rather than reproducing every line of algebra, we follow the "story of E" through each stage.

Target: Equation 11.38 — The Hamiltonian

$$\hat{H} = \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \left(\hat{a}^\dagger_{\mathbf{p}} \hat{a}_{\mathbf{p}} + \tfrac{1}{2}(2\pi)^3 \delta^{(3)}(\mathbf{0}) \right) \tag{2}$$

Step 0: The classical Hamiltonian density

The Hamiltonian density for a real scalar field of mass m is

$$\mathcal{H} = \frac{1}{2}\hat{\pi}^2 + \frac{1}{2}(\nabla\hat{\phi})^2 + \frac{1}{2}m^2\hat{\phi}^2,$$

with conjugate momentum $\hat{\pi} = \partial_0\hat{\phi}$. The full Hamiltonian is $\hat{H} = \int d^3x\, \mathcal{H}$.

***E*-status at Step 0**

No explicit E factors yet — they are hidden inside $\hat{\phi}$ and $\hat{\pi}$ via the mode expansion.

Step 1: Compute $\hat{\pi}$ from the mode expansion

Taking ∂_0 of the mode expansion (1):

$$\hat{\pi}(x) = \partial_0\hat{\phi} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \Big[(iE_{\mathbf{p}})\, \hat{a}_{\mathbf{p}}\, e^{ip\cdot x} + (-iE_{\mathbf{p}})\, \hat{a}^\dagger_{\mathbf{p}}\, e^{-ip\cdot x} \Big].$$

The time derivative pulls down $\pm iE_{\mathbf{p}}$.

***E*-status at Step 1**

Each mode in $\hat{\pi}$ now carries $\dfrac{E_{\mathbf{p}}}{\sqrt{2E_{\mathbf{p}}}} = \sqrt{\dfrac{E_{\mathbf{p}}}{2}}$. Compare with $\hat{\phi}$, whose modes carry $1/\sqrt{2E_{\mathbf{p}}}$. So:

Field	E-factor per mode
$\hat{\phi}$	$1/\sqrt{2E_{\mathbf{p}}}$
$\hat{\pi}$	$\sqrt{E_{\mathbf{p}}/2}$

Step 2: Compute the three quadratic terms

We need $\hat{\pi}^2$, $(\nabla\hat{\phi})^2$, and $m^2\hat{\phi}^2$. Each is a product of two mode expansions, so we integrate over *two* momenta $\mathbf{p}$ and $\mathbf{p}'$.

(i) $\hat{\pi}^2$: Two factors of $\hat{\pi}$, each contributing $\sqrt{E/2}$. The E-dependent prefactor from the two mode expansions is:

$$\sqrt{\frac{E_{\mathbf{p}}}{2}}\sqrt{\frac{E_{\mathbf{p}'}}{2}} = \frac{\sqrt{E_{\mathbf{p}}\,E_{\mathbf{p}'}}}{2}.$$

(ii) $(\nabla\hat{\phi})^2$: The gradient ∇ acting on $e^{\pm ip\cdot x}$ pulls down $\pm i\mathbf{p}$. Combined with the $1/\sqrt{2E}$ from $\hat{\phi}$, each mode contributes $|\mathbf{p}|/\sqrt{2E_{\mathbf{p}}}$. The product of two such factors gives:

$$\frac{|\mathbf{p}|\,|\mathbf{p}'|}{2\sqrt{E_{\mathbf{p}}\,E_{\mathbf{p}'}}}.$$

(iii) $m^2\hat{\phi}^2$: Straightforward — two factors of $1/\sqrt{2E}$:

$$\frac{m^2}{2\sqrt{E_{\mathbf{p}}\,E_{\mathbf{p}'}}}.$$

E-status at Step 2

At this stage the energy factors sit inside double momentum integrals and have not yet simplified.

Step 3: Do the spatial integral $\int d^3x$

The spatial integral acts on the plane-wave phases. Products of exponentials fall into two classes:

- **"Diagonal" terms** ($\hat{a}_{\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}'}$ and $\hat{a}^\dagger_{\mathbf{p}}\hat{a}_{\mathbf{p}'}$): phases are $e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}}$, so $\int d^3x$ yields $(2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{p}')$. This **collapses the double integral to a single integral** and sets $\mathbf{p}'=\mathbf{p}$, hence $E_{\mathbf{p}'}=E_{\mathbf{p}}$.
- **"Off-diagonal" terms** ($\hat{a}_{\mathbf{p}}\hat{a}_{\mathbf{p}'}$ and $\hat{a}^\dagger_{\mathbf{p}}\hat{a}^\dagger_{\mathbf{p}'}$): phases are $e^{i(\mathbf{p}+\mathbf{p}')\cdot\mathbf{x}}$, so $\int d^3x$ yields $(2\pi)^3\delta^{(3)}(\mathbf{p}+\mathbf{p}')$, setting $\mathbf{p}'=-\mathbf{p}$, hence again $E_{\mathbf{p}'}=E_{\mathbf{p}}$.

E-status at Step 3

After the delta functions collapse one integral, every energy factor involves only $E_{\mathbf{p}}$. The prefactors simplify dramatically:

Term	Before δ	After δ
$\hat{\pi}^2$	$\frac{\sqrt{E_{\mathbf{p}}\,E_{\mathbf{p}'}}}{2}$	$\frac{E_{\mathbf{p}}}{2}$
$(\nabla\hat{\phi})^2$	$\frac{\lvert\mathbf{p}\rvert\,\lvert\mathbf{p}'\rvert}{2\sqrt{E_{\mathbf{p}}\,E_{\mathbf{p}'}}}$	$\frac{\lvert\mathbf{p}\rvert^2}{2E_{\mathbf{p}}}$
$m^2\hat{\phi}^2$	$\frac{m^2}{2\sqrt{E_{\mathbf{p}}\,E_{\mathbf{p}'}}}$	$\frac{m^2}{2E_{\mathbf{p}}}$

Step 4: Cancel the off-diagonal terms

The off-diagonal ($\hat{a}\hat{a}$ and $\hat{a}^\dagger\hat{a}^\dagger$) terms come with time-dependent phases $e^{\pm 2iE_{\mathbf{p}}t}$. Let us examine their E-dependent coefficients. For $\hat{a}_{\mathbf{p}}\hat{a}_{-\mathbf{p}}$:

- From $\frac{1}{2}\hat{\pi}^2$: coefficient is $+\frac{1}{2}\cdot\left(-\frac{E_{\mathbf{p}}}{2}\right) = -\frac{E_{\mathbf{p}}}{4}$ (the minus sign comes from the $(iE)(iE) = -E^2$ in $\hat{\pi}$, divided by $2E$).
- From $\frac{1}{2}(\nabla\hat{\phi})^2$: coefficient is $+\frac{1}{2}\cdot\frac{|\mathbf{p}|^2}{2E_{\mathbf{p}}}$.
- From $\frac{1}{2}m^2\hat{\phi}^2$: coefficient is $+\frac{1}{2}\cdot\frac{m^2}{2E_{\mathbf{p}}}$.

Adding the last two:

$$\frac{|\mathbf{p}|^2 + m^2}{4E_{\mathbf{p}}} = \frac{E_{\mathbf{p}}^2}{4E_{\mathbf{p}}} = \frac{E_{\mathbf{p}}}{4}.$$

This *exactly cancels* the $-E_{\mathbf{p}}/4$ from the $\hat{\pi}^2$ term!

E-status at Step 4 — the crucial cancellation

The **dispersion relation** $E_{\mathbf{p}}^2 = |\mathbf{p}|^2 + m^2$ causes the off-diagonal terms to vanish:

$$\underbrace{-\frac{E_{\mathbf{p}}}{4}}_{\text{from } \hat{\pi}^2} + \underbrace{+\frac{|\mathbf{p}|^2 + m^2}{4E_{\mathbf{p}}}}_{\text{from } (\nabla\hat{\phi})^2 + m^2\hat{\phi}^2} = -\frac{E_{\mathbf{p}}}{4} + \frac{E_{\mathbf{p}}}{4} = 0.$$

All time-dependent ($e^{\pm 2iEt}$) pieces drop out, as they must for a time-independent Hamiltonian.

Step 5: Collect the diagonal terms

Only the $\hat{a}^\dagger\hat{a}$ and $\hat{a}\hat{a}^\dagger$ terms survive. Their energy prefactors *add* rather than cancel:

- From $\frac{1}{2}\hat{\pi}^2$: $+\frac{E_{\mathbf{p}}}{4}$.
- From $\frac{1}{2}(\nabla\hat{\phi})^2 + \frac{1}{2}m^2\hat{\phi}^2$: $+\frac{|\mathbf{p}|^2+m^2}{4E_{\mathbf{p}}} = +\frac{E_{\mathbf{p}}}{4}$.

Sum: $\frac{E_{\mathbf{p}}}{4} + \frac{E_{\mathbf{p}}}{4} = \frac{E_{\mathbf{p}}}{2}$.

Including both operator orderings ($\hat{a}\hat{a}^\dagger$ and $\hat{a}^\dagger\hat{a}$), the integrand is:

$$\frac{E_{\mathbf{p}}}{2}\left(\hat{a}_{\mathbf{p}}\hat{a}_{\mathbf{p}}^\dagger + \hat{a}_{\mathbf{p}}^\dagger\hat{a}_{\mathbf{p}}\right).$$

Using $\hat{a}_{\mathbf{p}}\hat{a}_{\mathbf{p}}^\dagger = \hat{a}_{\mathbf{p}}^\dagger\hat{a}_{\mathbf{p}} + (2\pi)^3\delta^{(3)}(\mathbf{0})$:

$$= \frac{E_{\mathbf{p}}}{2}\left(2\hat{a}_{\mathbf{p}}^\dagger\hat{a}_{\mathbf{p}} + (2\pi)^3\delta^{(3)}(\mathbf{0})\right) = E_{\mathbf{p}}\left(\hat{a}_{\mathbf{p}}^\dagger\hat{a}_{\mathbf{p}} + \tfrac{1}{2}(2\pi)^3\delta^{(3)}(\mathbf{0})\right).$$

E-status at Step 5 — arrival at Eq. 11.38

The two $E_{\mathbf{p}}/4$ contributions *add* to give $E_{\mathbf{p}}/2$, and after combining operator orderings we get a single factor of $\boldsymbol{E_{\mathbf{p}}}$ multiplying the number operator plus the zero-point piece.

Step 6: The final result

Equation 11.38 — The Hamiltonian

$$\hat{H} = \int \frac{d^3p}{(2\pi)^3}\, E_{\mathbf{p}} \left(\hat{a}^\dagger_{\mathbf{p}} \hat{a}_{\mathbf{p}} + \frac{1}{2}(2\pi)^3 \delta^{(3)}(\mathbf{0}) \right)$$

Physical interpretation: The Hamiltonian is a sum over all momentum modes, each contributing energy $E_{\mathbf{p}}$ times the number of particles in that mode (plus an infinite zero-point energy that is removed by normal ordering).

Summary: The Complete Story of E

Mode expansion: Each mode carries $1/\sqrt{2E_{\mathbf{p}}}$ from Lorentz-invariant normal-

Commutator & cleanup: $\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} = 2\hat{a}^\dagger\hat{a} + [\hat{a}, \hat{a}^\dagger]$ gives final factor of $E_{\mathbf{p}}$.

The $1/\sqrt{2E}$ in the mode expansion might look like it would produce $1/E$ in the Hamiltonian. Instead, the time derivative injects one power of E (turning $1/\sqrt{2E}$ into $\sqrt{E/2}$), the dispersion relation $E^2 = |\mathbf{p}|^2 + m^2$ kills the unwanted terms and doubles the wanted ones, and the algebra conspires to produce a clean, intuitive result: each mode contributes energy $E_{\mathbf{p}}$ times the number of quanta it contains.

End of solution.